

OpenSeaMap

the free

Nautical Chart

71% Sea


ATLANTISCHER
PAZIFISCHER
OZEAN
INDISCHER
OZEAN
OZEAN
ANTARKTISCHER OZEAN







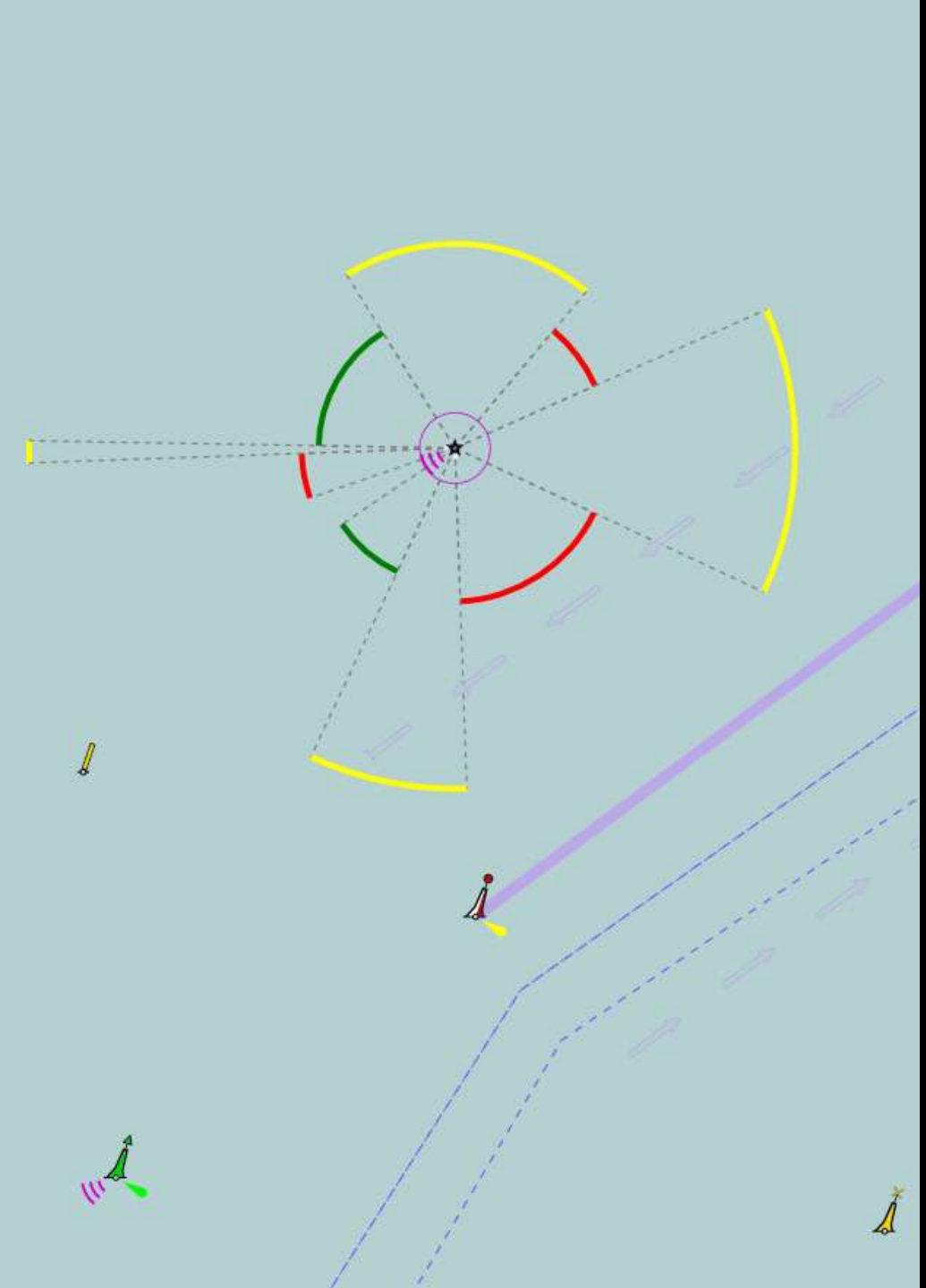

BALU





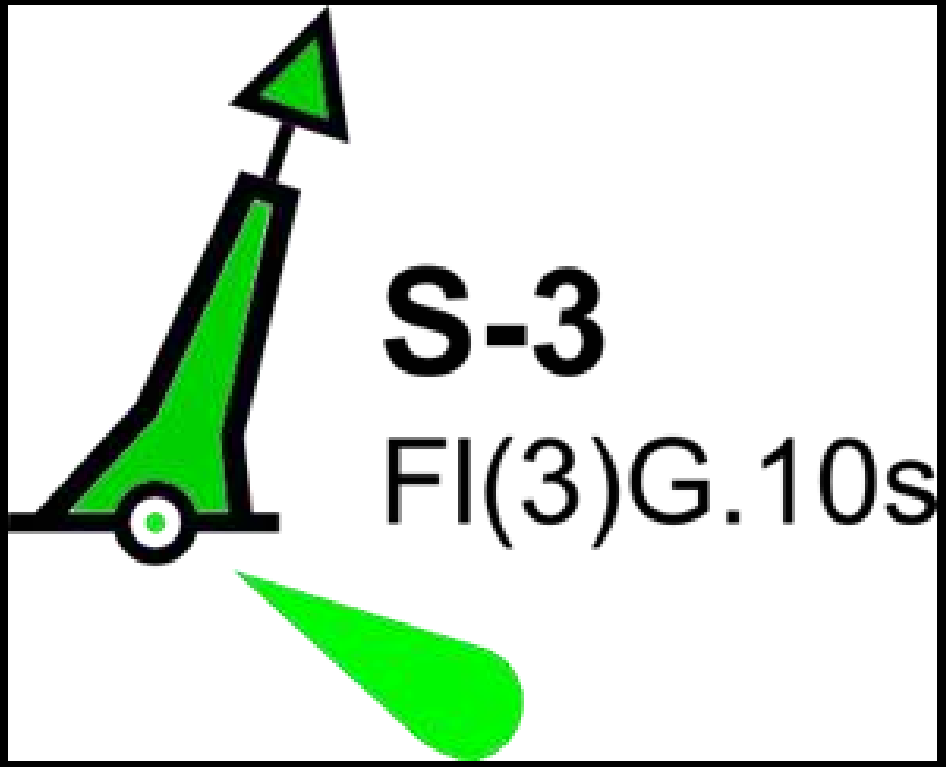




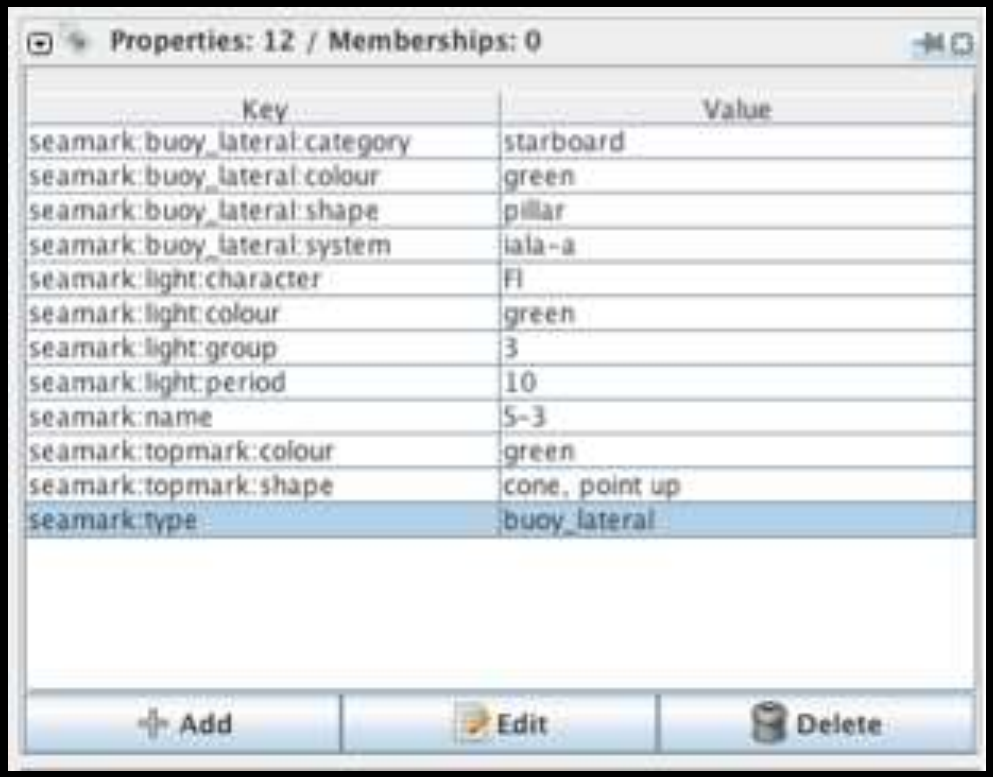


JOSM
SeaMap Editor

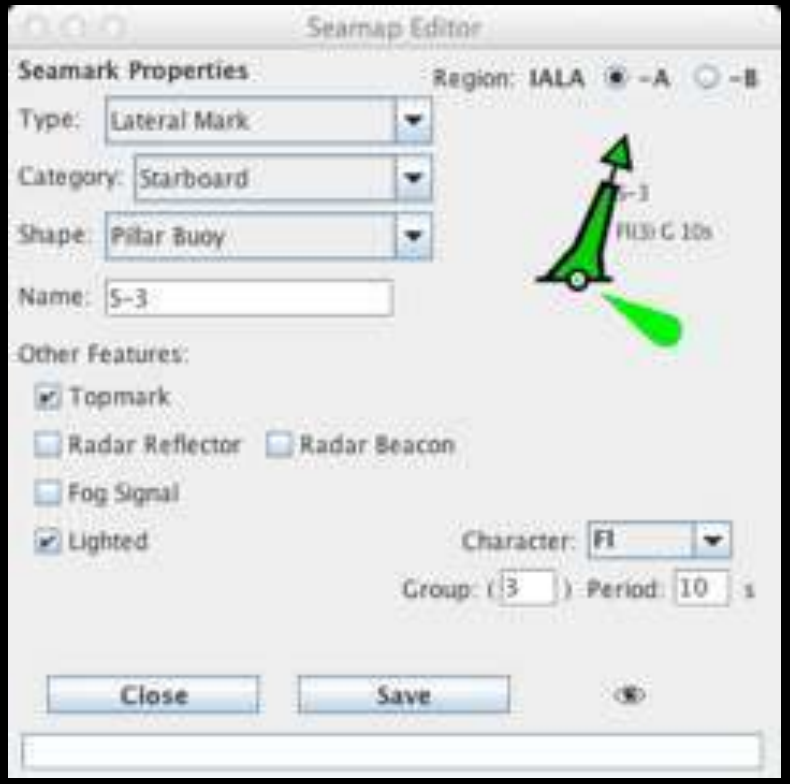
We will see this



Complex data scheme



Current JOSM plugin



User-friendly GUI



A lot of Information…



Seamark

A simple buoy...

- Function



Seamark

A simple buoy…

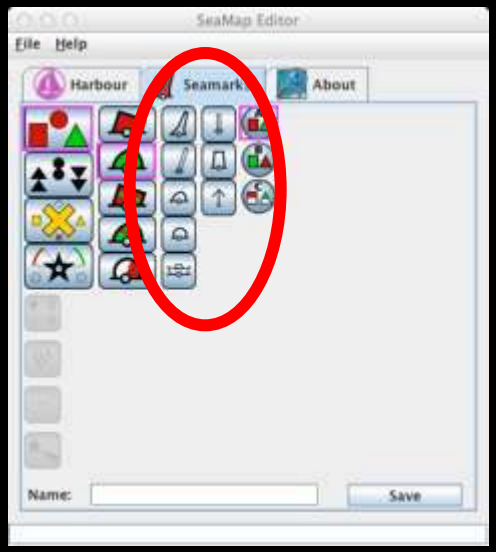
- Function
- Category



Seamark

A simple buoy…

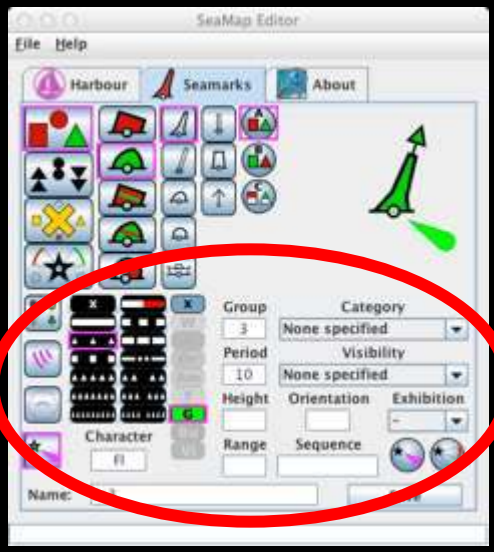
- Function
- Category
- Form



Seamark

A simple buoy…

- Function
- Category
- Form
- Topmark, Light, …





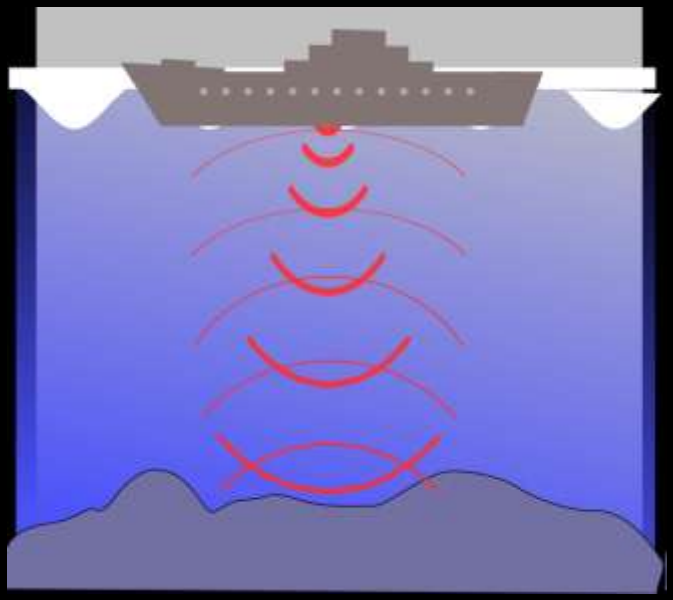
project @ OpenSeaMap.org





OpenSeaMap
and
Deep Water

GEBCO for OpenSeaMap



Global Dataset

Release 2009

30“ Grid ~500 m

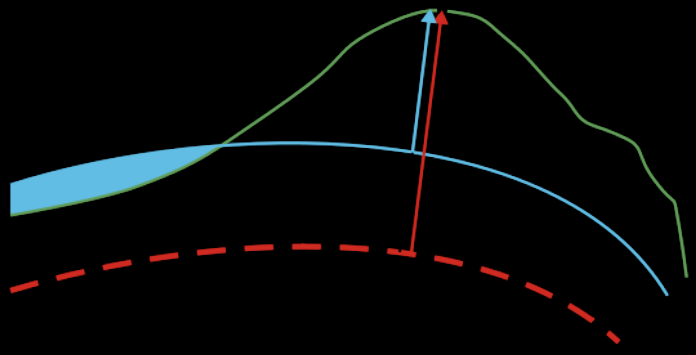
290 Million soundings

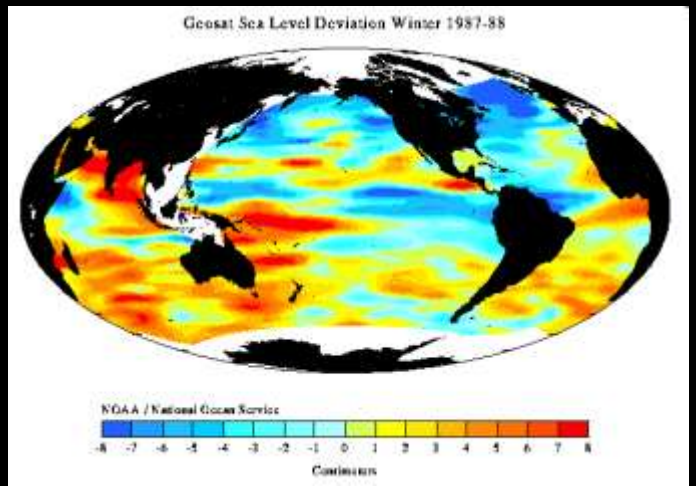
1 Billion Points

Agreement
with OpenSeaMap

Sea Level

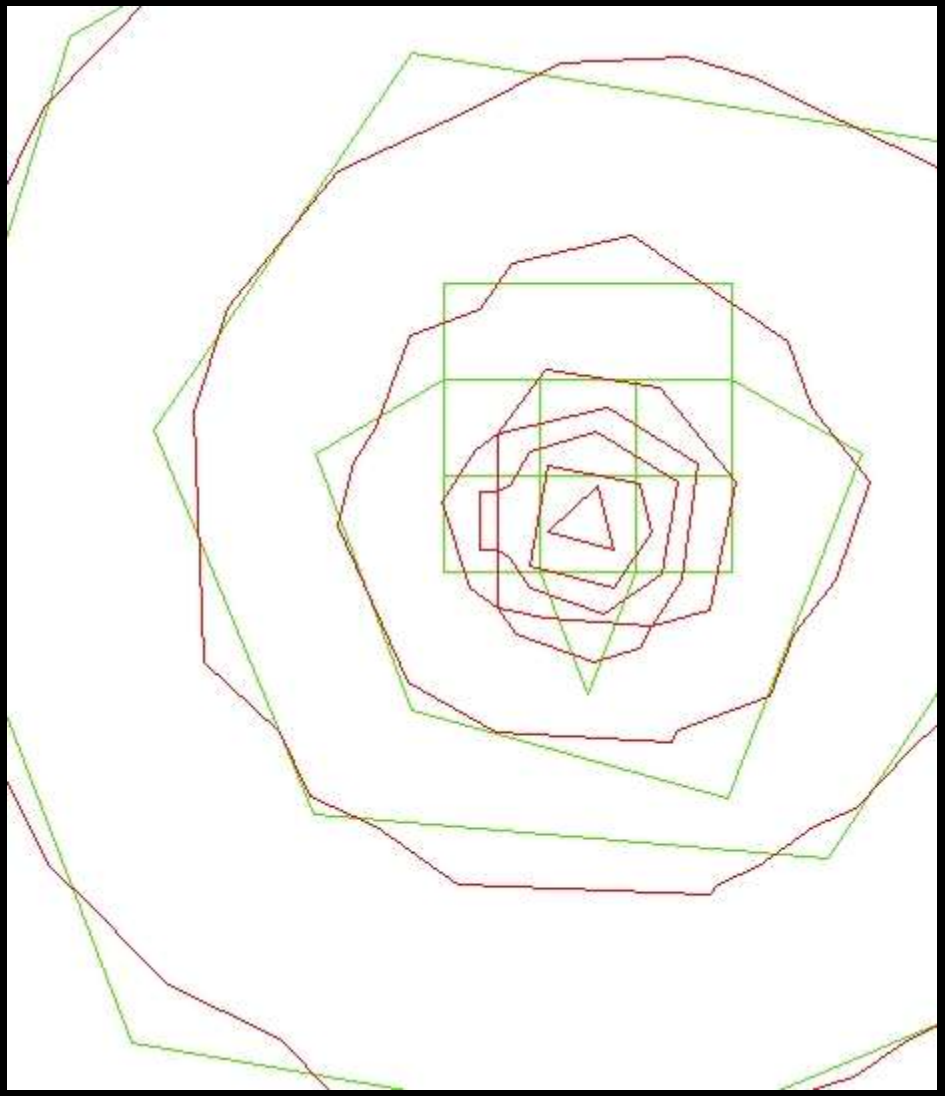





Geosat Sea Level Deviation Winter 1987-88
NOAA / National Ocean Service
-8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8
Centimeters

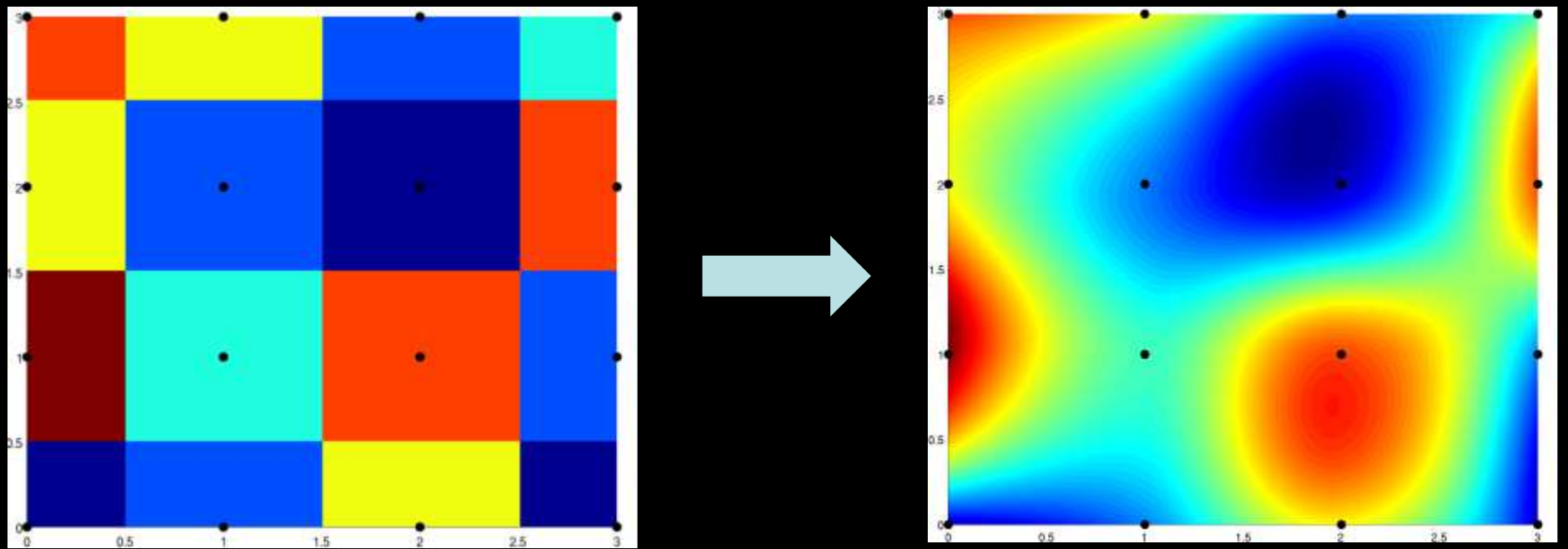

250-500
500-750
750-1000
1000-1250
1250-1500
1500-2000
2000-2500
2500-3000
3000-3500
3500-4000
4000-4500
4500-5000
5000-5500
5500-6000
6000-6500
6500-7000
7000-7500
7500-8000
8000-8500
8500-9000
9000-9500
9500- ...

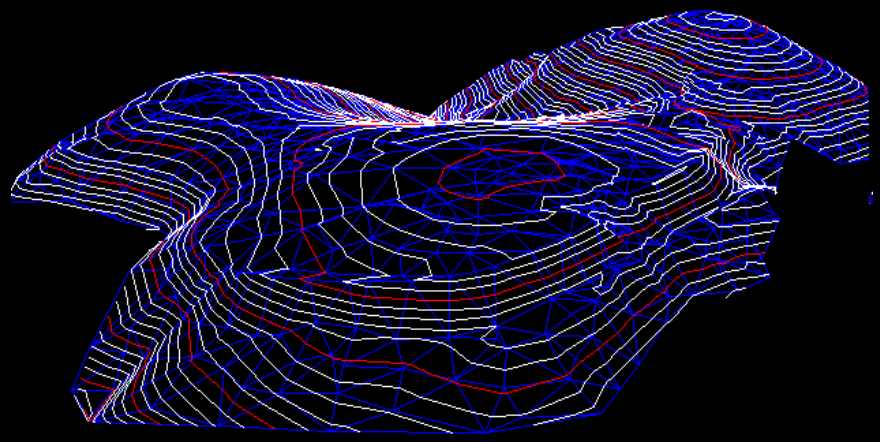
23 Classes



Not nice…

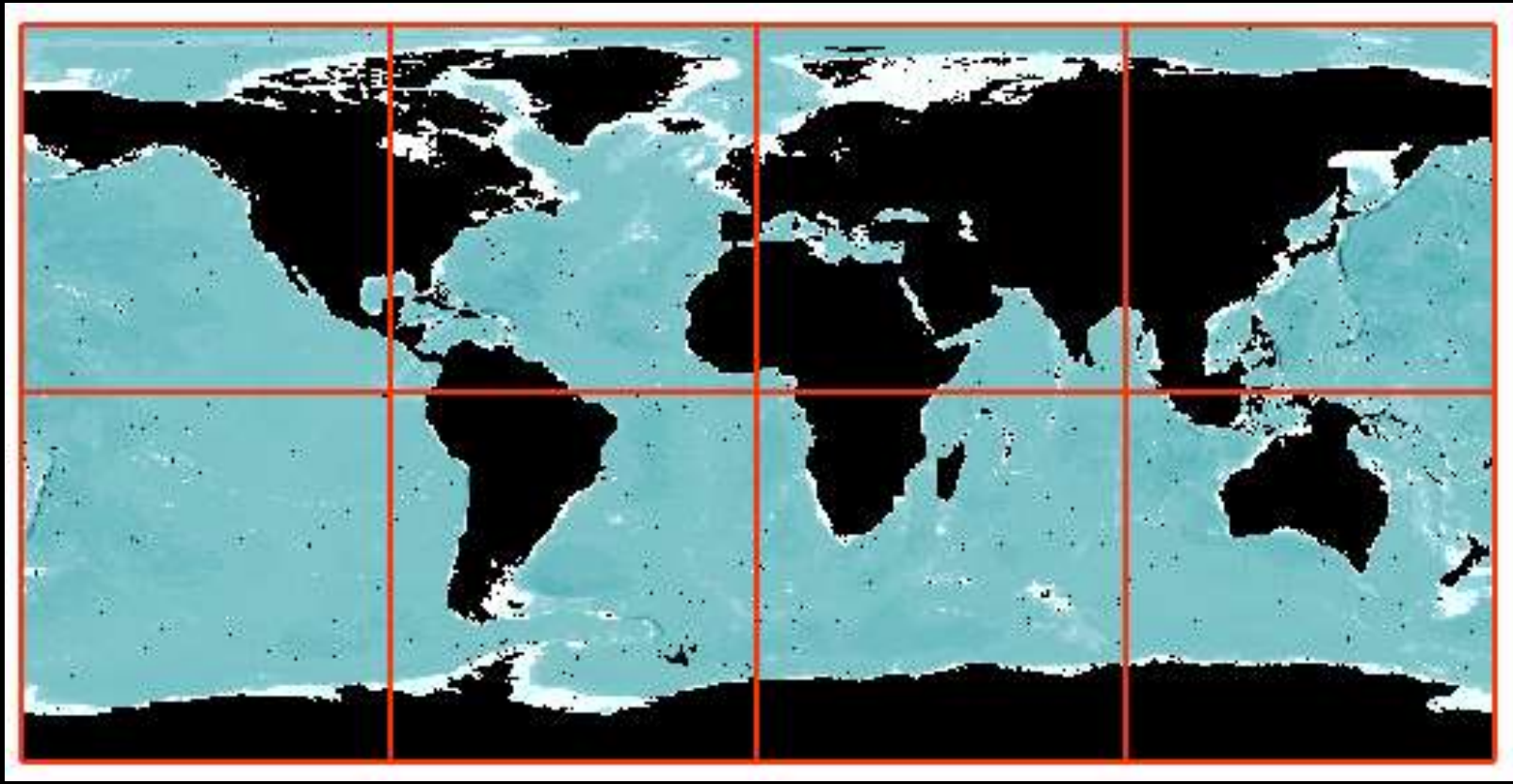
Bicubic Interpolation



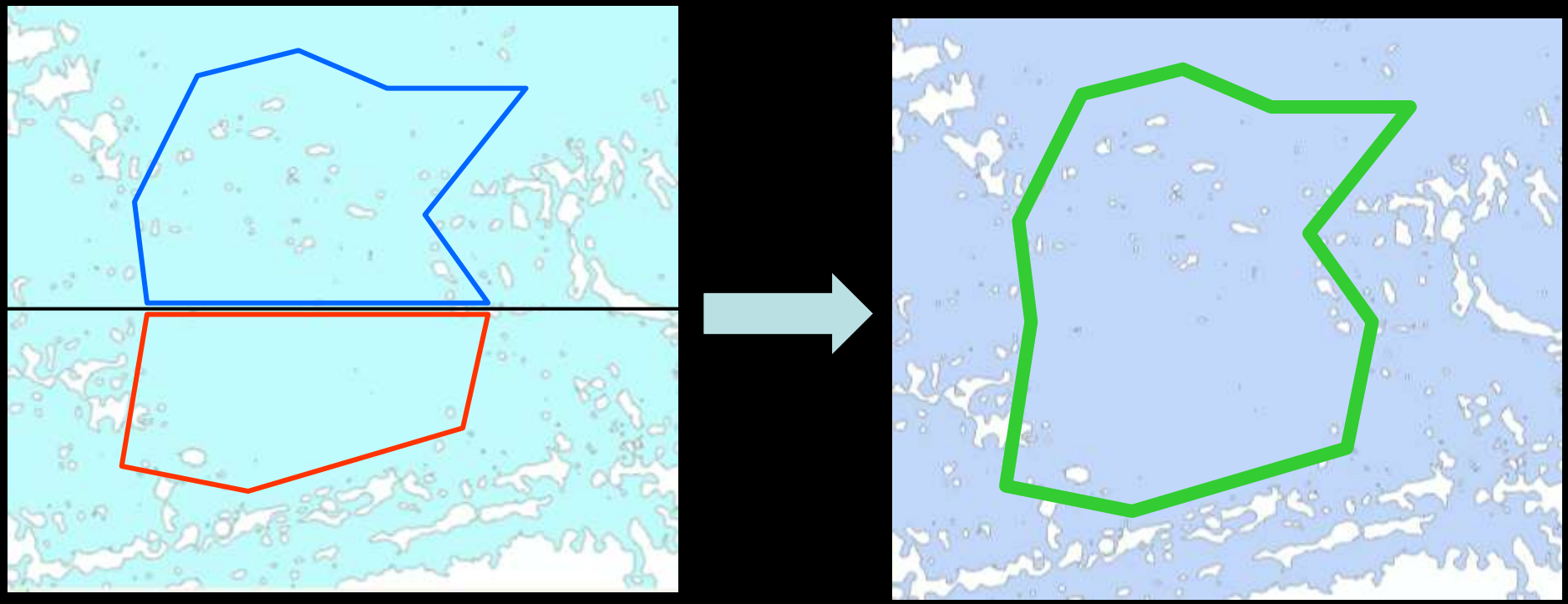


Nice…
But too big

Split



Resolving of Intersection

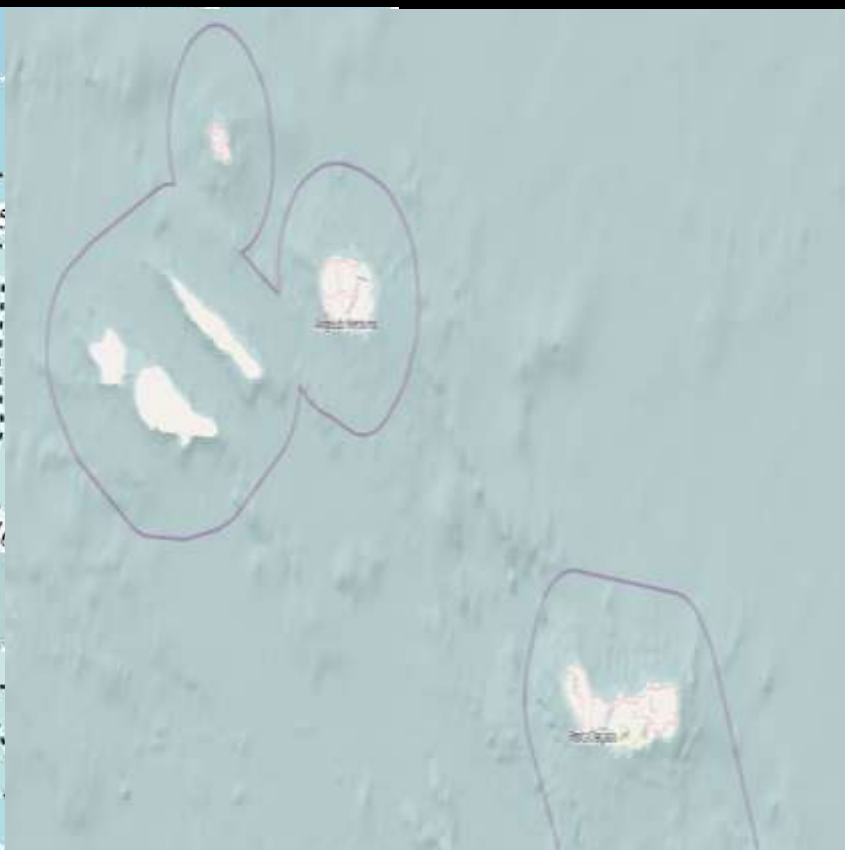


Polygon

Polyline

Sea-Shade








ESRI
ArcGIS

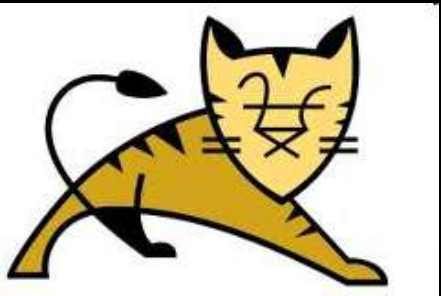

PostGIS
Spatial PostgreSQL


Ps


debian




GeoWebCache




GeoServer



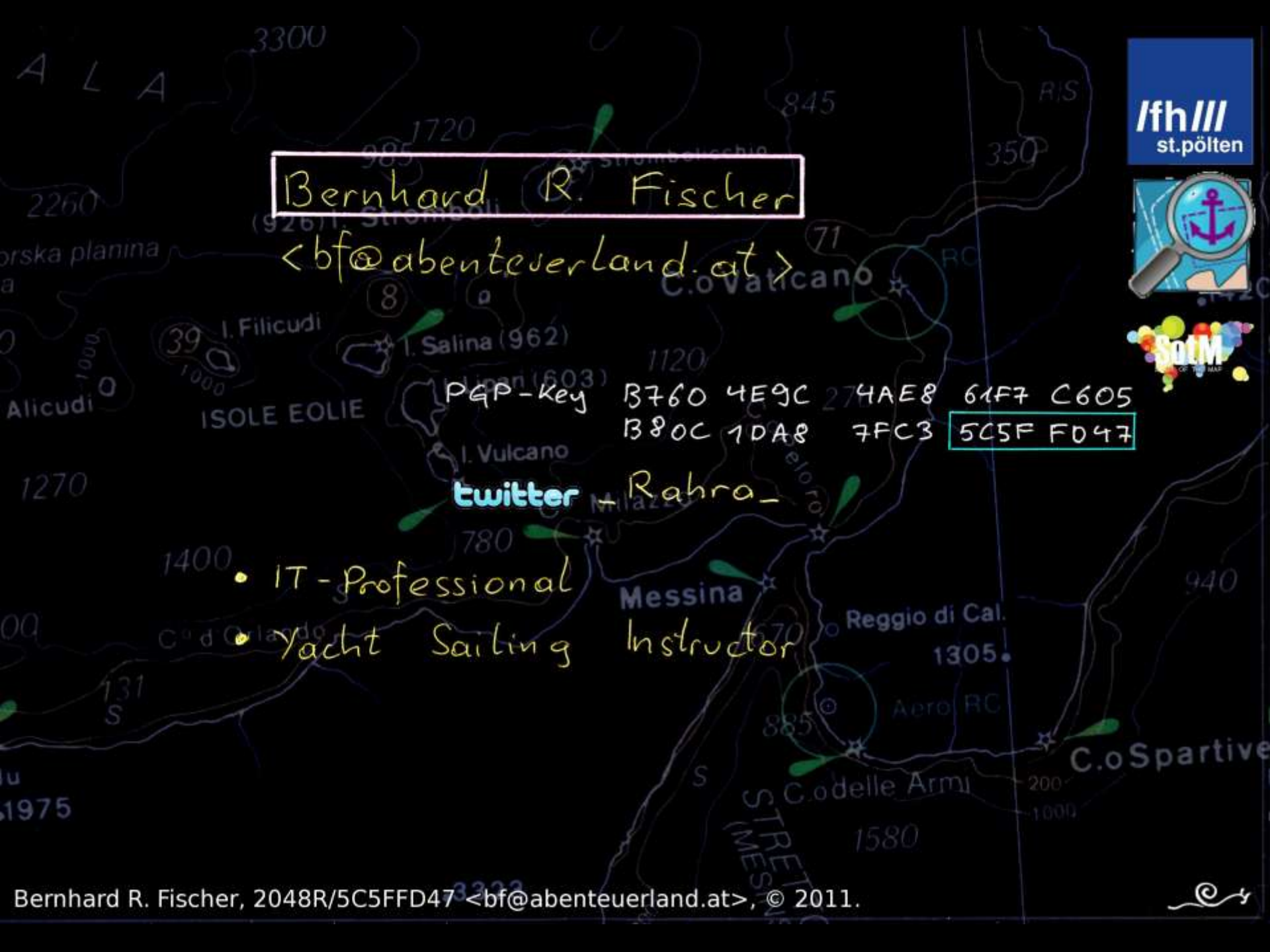
Help wanted:

- Faster tile server
- Cutting land without opacity

project @ OpenSeaMap.org



Import of Lights



Bernhard R. Fischer, 2048R/5C5FFD47 <bf@abenteuerland.at>, © 2011.

Marine Lights and Seamarks







Very important for navigation of vessels.

Bernhard R. Fischer, 2048R/5C5FFD47 <bf@abenteuerland.at>, © 2011.

Mapping of seamarks

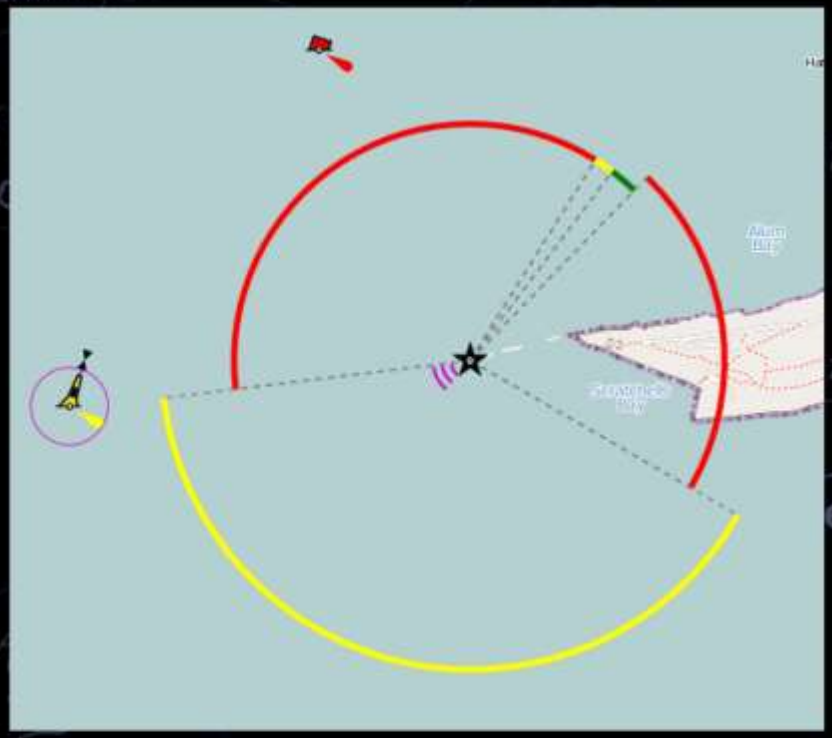
Roadmap: man_made=lighthouse



In a Seachart: many different attributes

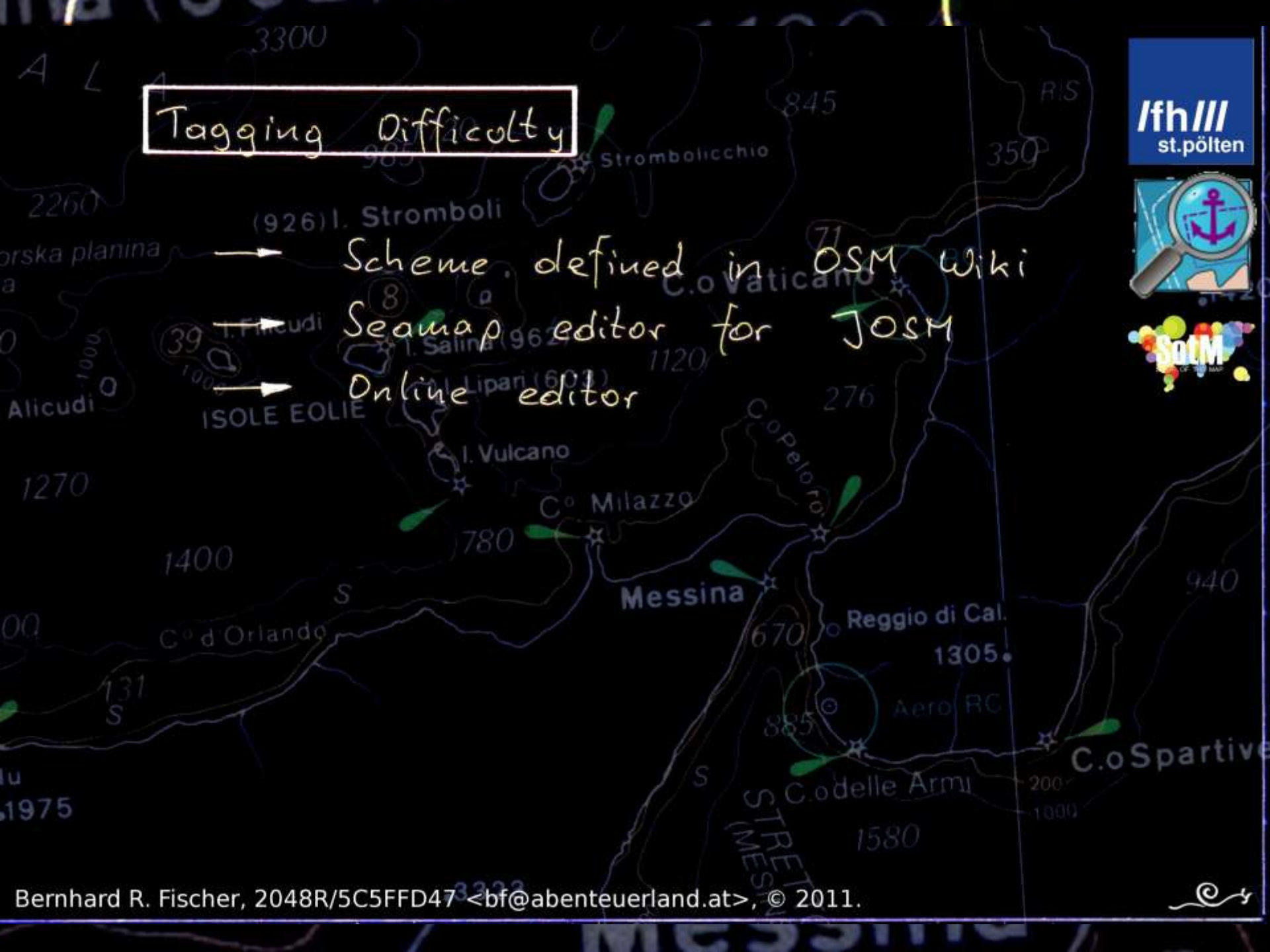
Bernhard R. Fischer, 2048R/5C5FFD47 <bf@abenteuerland.at>, © 2011.

The Needles Lighthouse



```
man_made = lighthouse
name = The Needles Lighthouse
seamark = lighthouse
seamark:fog_signal:category = horn
seamark:fog_signal:group = 2
seamark:fog_signal:period = 30
seamark:light:1 = red:224:300:140
seamark:light:1:colour = red
seamark:light:1:sector_end = 300
seamark:light:1:sector_start = 224
seamark:light:2 = white:300:083:170
seamark:light:2:colour = white
seamark:light:2:sector_end = 083
seamark:light:2:sector_start = 300
seamark:light:3 = red:083:212:130
seamark:light:3:colour = red
seamark:light:3:sector_end = 212
seamark:light:3:sector_start = 083
seamark:light:4 = white:212:217:130
seamark:light:4:colour = white
seamark:light:4:sector_end = 217
seamark:light:4:sector_start = 212
seamark:light:5 = green:217:224:130
seamark:light:5:colour = green
seamark:light:5:sector_end = 224
seamark:light:5:sector_start = 217
seamark:light:character = Oc
seamark:light:group = 2
seamark:light:height = 24
seamark:light:period = 20
seamark:light:range = 15
seamark:name = Needles
seamark:type = landmark
source = OS OpenData StreetView
wikipedia = en:The_Needles
```

Bernhard R. Fischer, 2048R/5C5FFD47 <bf@abenteuerland.at>, © 2011.


Tagging Difficulty
Scheme defined in OSM Wiki
Seamap editor for JOSM
Online editor
lfh
st.pölten
SotM
Strombolicchio
(926) I. Stromboli
ISOLE EOLIE
Alicudi
I. Vulcano
C° Milazzo
C.o Vaticano
C.o Peloro
Messina
C° d'Orlando
Reggio di Cal.
1305
Aero RC
C.o delle Armi
C.o Spartivento
STRETTO DI MESSINA
Bernhard R. Fischer, 2048R/5C5FFD47 <bf@abenteuerland.at>, © 2011.

Where do those Lights come from?

- Manual survey
  - (+) position correct
  - (−) attributes doubtful
- List of Lights (LoL)
  - ~ 50k officially IALA-registered Lights
  - automatic import





584 **-Needles.** A 0528 — 50° 39.7′ N, 1° 35.4′ W — **Oc.(2)W.R.G.** period 20s, lt. 2s, ec. 2s, lt. 14s, ec. 2s — 80 W. **17** R. **24** R. 14 G. 14 — Round granite tower, red band, red lantern; 102. Helicopter platform. — R. shore-300°, W.-083°, R. (unintensified)-212°, W.-217°, G.-224°. **Horn:** 2 bl. ev. 30s.

Bernhard R. Fischer, 2048R/5C5FFD47 <bf@abenteuerland.at>, © 2011.

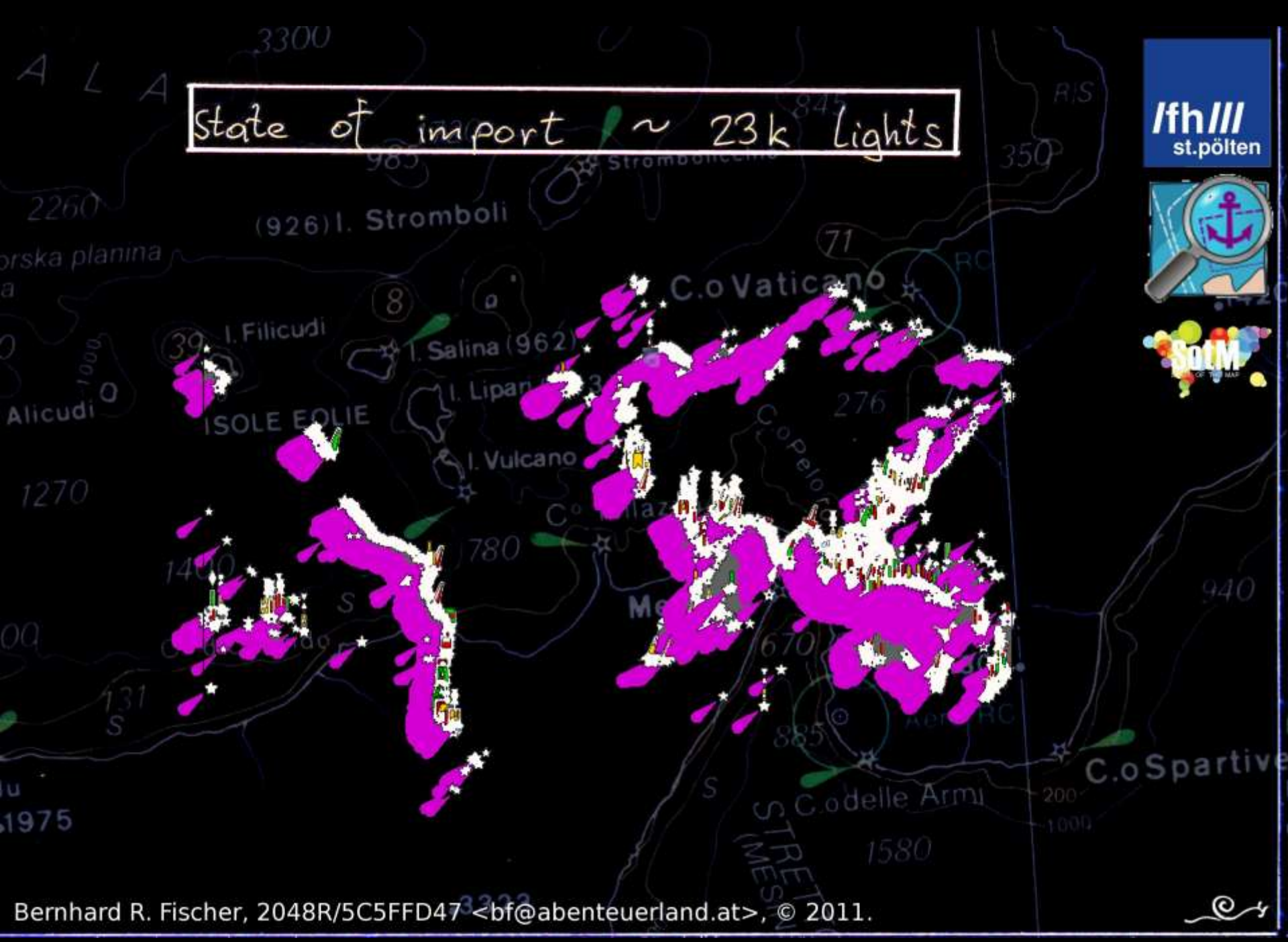
Attributes vs. Position accuracy

- Controversal opinions

→ Positions may be corrected easily
but attributes are not

high zoom level != high pos. accuracy

Bernhard R. Fischer, 2048R/5C5FFD47 <bf@abenteuerland.at>, © 2011.



Bernhard R. Fischer, 2048R/5C5FFD47 <bf@abenteuerland.at>, © 2011.

Problems

- Maintenance
- Consistency of reference IDs
- LoL updates





Bernhard R. Fischer, 2048R/5C5FFD47 <bf@abenteuerland.at>, © 2011.



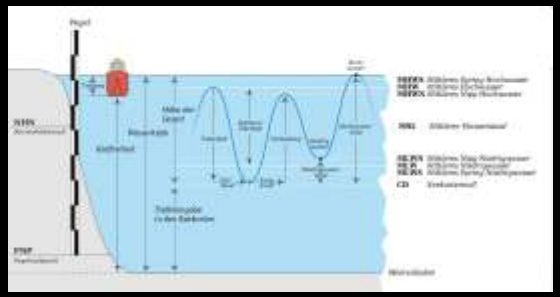
Visions













mapnik

